

DEPARTMENT OF ELECTRONICS ENGINEERING

ELE366

COMMUNICATIONS LABORATORY

Single Side Band Modulation&Demodulation

Frequency Modulation&Demodulation

EXPERIMENT-2

1. GOAL

In this experiment, you will create a an Single Sideband (SSB) signal. As well you will again use product demodulation to recover the message and examine the effect of phase and frequency errors on the recovery process. Afterwards, you will learn how to perform the frequency modulation and demodulation.

2. BACKGROUND

2.1. Single Side-band Modulation & Demodulation

Single side band (SSB) modulation is a suppressed carrier and suppressed lower (or upper) side band amplitude modulation. Let the message $m(t)$ be a band-limited signal as

$$m(t) = A_m cos(2\pi f_m t)$$

where, A_m is the amplitude of the message signal and f_m is the message frequency. The generation of the message signal is illustrated in Figure 2.1.3. An SSB modulated signal may thus be described as a function of time as follows:

$$y_{SSB}(t) = A_m sin(2\pi f_m t) V_c sin(2\pi f_c t) + A_m cos(2\pi f_m t) V_c cos(2\pi f_c t)$$

For more clarity,

$$y_{SSB}(t) = A_m V_c cos(2\pi(f_c + f_m)t)$$

A description of the modulation process for the suppressed upper band is illustrated in Figure 2.1.1. An alternative description of the modulation process is illustrated in Figure 2.1.2.

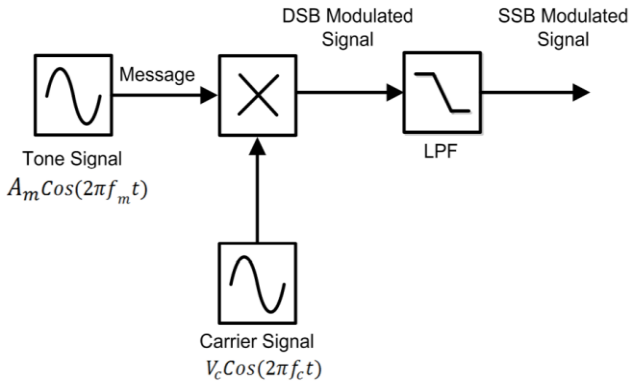


Figure 2.1.1. Suppressed upper band SSB modulated signal generation

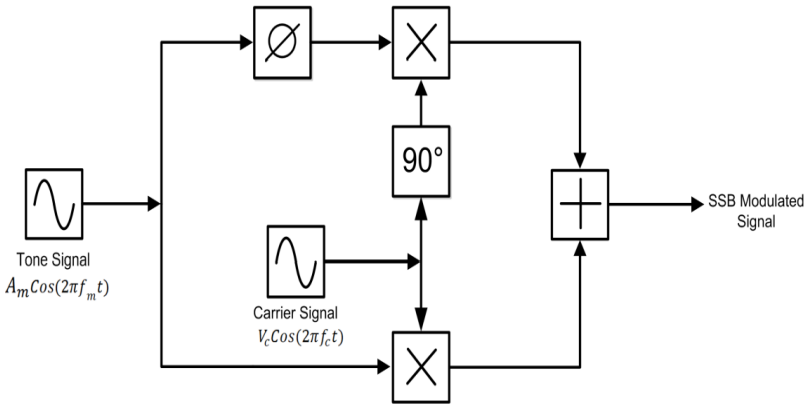


Figure 2.1.2. Alternative SSB modulated signal generation

Demodulation of an SSB signal can be accomplished in two steps: down-conversion and low pass filtering. We can assume that transmitted $m(t)$ is a band-limited signal. Additionally, we assume that the receiver knows carrier frequency (f_c) accurately. We first generate a down-converted base-band signal as follows:

$$D(t) = y_{SSB}(t)cos(2\pi f_c t)$$

For more clarity,

$$D(t) = V_c \frac{A_m}{2} cos(2\pi f_m t) + V_c \frac{A_m}{2} cos(2\pi f_m t)cos(4\pi f_c t)$$

$$D(t) = V_c \frac{A_m}{2} cos(2\pi f_m t) + V_c \frac{A_m}{4} cos(2\pi(2f_c - f_m)t) + V_c \frac{A_m}{4} cos(2\pi(2f_c + f_m)t)$$

After proper low-pass filtering, we can get the recovered message signal with an amplitude scaled by 0.5.

$$\hat{m}(t) = V_c \frac{A_m}{2} cos(2\pi f_m t)$$

The conventional scheme for demodulation of SSB modulated signal is a circuit with a mixer and a proper low-pass filter. This scheme is shown in Figure 2.1.3. As a disadvantage of SSB modulation, carrier tone must be recovered accurately with carrier frequency (f_c) and carrier phase as in transmitter unit.

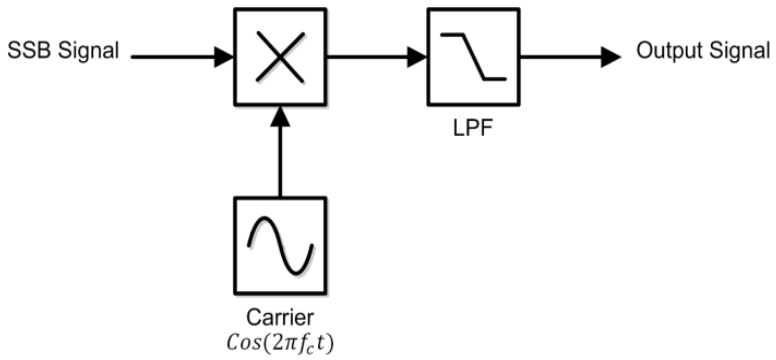


Figure 2.1.3. Demodulation of an SSB modulated signal

2.2. Frequency Modulation & Demodulation

Frequency Modulation (FM) controls the carrier frequency by using the message signal. Let the message m(t) be a band-limited signal. The frequency modulated signal is considered as follows:

$$y_{FM}(t) = V_c \cos\left(2\pi f_c t + k_f \int_0^t m(\tau)d\tau\right)$$

where, V_c is the amplitude of the modulated signal and f_c is the carrier frequency and k_f is the frequency sensibility. Alternatively, phase modulation (PM) is described as follows:

$$y_{PM}(t) = V_c \cos\left(2\pi f_c t + k_f m(t)\right)$$

Let us consider the message signal as given below

$$m(t) = A_m cos(2\pi f_m t)$$

The generation of the message signal is illustrated in Figure 2.2.1. Notice that there is not a DC component in the message signal. Then, frequency modulated signal may thus be described as a function of time as follows:

$$y_{FM}(t) = V_c \cos\left(2\pi f_c t + A_m \frac{k_f}{2\pi f_m} sin(2\pi f_m t)\right)$$

For more clarity,

$$y_{FM}(t) = V_c \cos\left(2\pi f_c t + \beta sin(2\pi f_m t)\right)$$

where, $\beta = \frac{\Delta_f}{f_m}$ is the frequency modulation index and $\Delta_f = \frac{A_m k_f}{2\pi}$ is the frequency deviation. A description of the modulation process is illustrated in Figure 2.2.2.

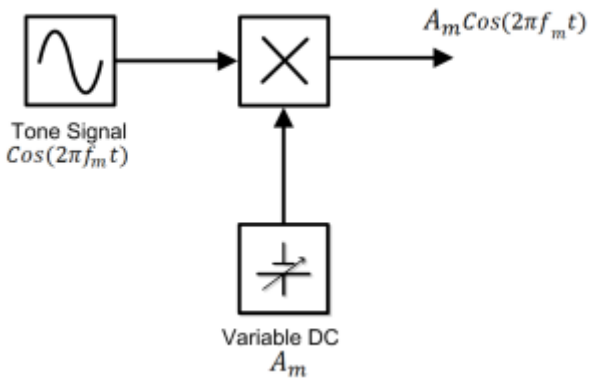


Figure 2.2.1. A simple message signal generation

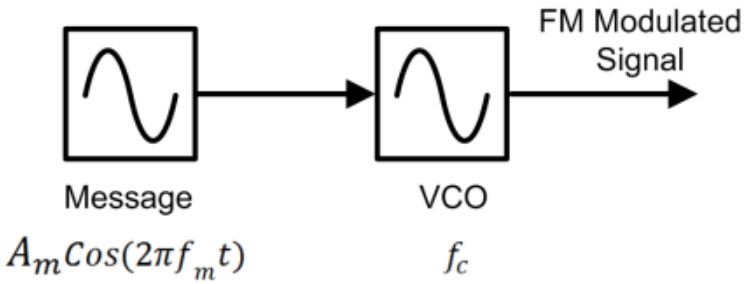


Figure 2.2.2. Frequency modulated signal generation

Let us consider a simple message signal with parameters Am = 1 V and fm = 5 kHz and carrier signal with fc = 100 kHz and Vc = 1 V. Frequency modulated signal is shown in Figure 2.2.3. In order to compute the frequency spectrum of the generated FM signal, Bessel functions are used to evaluate the serial expression of the modulated signal as follows:

$$y_{FM}(t) = V_c \cos\left(2\pi f_c t + \beta sin(2\pi f_m t)\right)$$

$$y_{FM}(t) = V_c J_0(\beta)\cos(2\pi f_c t) + V_c \sum_{k=0}^{\infty} J_{2k}(\beta)\left(\sin(2\pi f_c t + 2k f_m t) - \sin\left(2\pi f_c t - 2k f_m t\right)\right)$$

$$+ V_c \sum_{k=0}^{\infty} J_{2k+1}(\beta)\left(\sin(2\pi f_c t + (2k+1) f_m t) - \sin\left(2\pi f_c t - (2k+1) f_m t\right)\right)$$

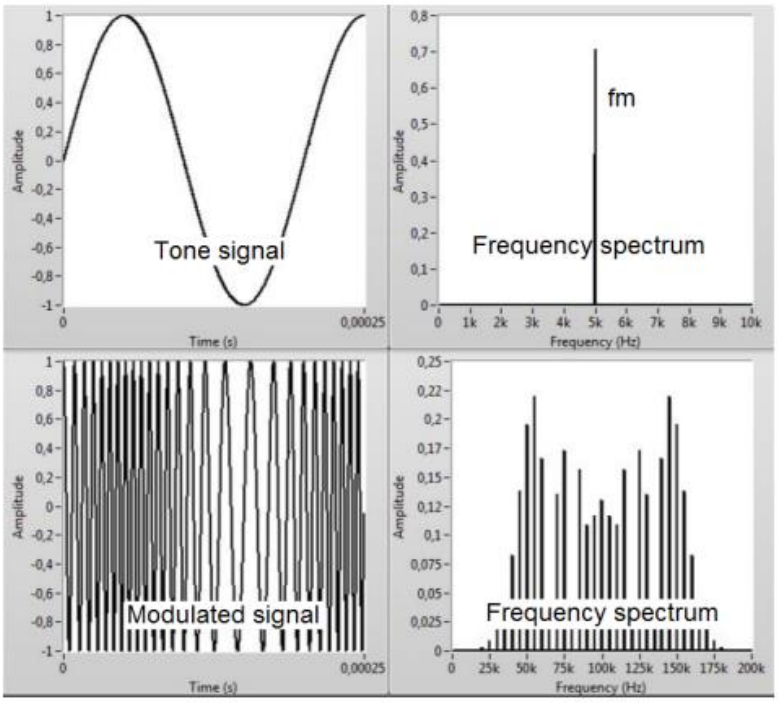


Figure 2.2.3. Frequency modulated signal in time and frequency domains

Bessel function is given in Table 2.1 for a given frequency modulation index β. In fact, number of frequency components goes to infinity. Practically, we can take a countable set of them (with %98 of the total power) to consider the frequency spectrum of FM signals. For this purpose, Carson Bandwidth Rule provide a bandwidth as follows:

$$B \cong 2(f_m + \Delta_f)$$

As an advantage of frequency modulation, transmission power is constant. In this way, the carrier amplitude is less important than AM and DSB and SSB modulation techniques. In fact, electrical noise is more effective on the carrier amplitude than the carrier frequency and hence FM performs higher transmission quality.

Modulation index	Sideband																
	Carrier	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0.00	1.00																
0.25	0.98	0.12															
0.5	0.94	0.24	0.03														
1.0	0.77	0.44	0.11	0.02													
1.5	0.51	0.56	0.23	0.06	0.01												
2.0	0.22	0.58	0.35	0.13	0.03												
2.41	0	0.52	0.43	0.20	0.06	0.02											
2.5	−0.05	0.50	0.45	0.22	0.07	0.02	0.01										
3.0	−0.26	0.34	0.49	0.31	0.13	0.04	0.01										
4.0	−0.40	−0.07	0.36	0.43	0.28	0.13	0.05	0.02									
5.0	−0.18	−0.33	0.05	0.36	0.39	0.26	0.13	0.05	0.02								
5.53	0	−0.34	−0.13	0.25	0.40	0.32	0.19	0.09	0.03	0.01							
6.0	0.15	−0.28	−0.24	0.11	0.36	0.36	0.25	0.13	0.06	0.02							
7.0	0.30	0.00	−0.30	−0.17	0.16	0.35	0.34	0.23	0.13	0.06	0.02						
8.0	0.17	0.23	−0.11	−0.29	−0.10	0.19	0.34	0.32	0.22	0.13	0.06	0.03					
8.65	0	0.27	0.06	−0.24	−0.23	0.03	0.26	0.34	0.28	0.18	0.10	0.05	0.02				
9.0	−0.09	0.25	0.14	−0.18	−0.27	−0.06	0.20	0.33	0.31	0.21	0.12	0.06	0.03	0.01			
10.0	−0.25	0.04	0.25	0.06	−0.22	−0.23	−0.01	0.22	0.32	0.29	0.21	0.12	0.06	0.03	0.01		
12.0	0.05	−0.22	−0.08	0.20	0.18	−0.07	−0.24	−0.17	0.05	0.23	0.30	0.27	0.20	0.12	0.07	0.03	0.01

Table 2.1. : Bessel functions (J_k) for a given (β)

To demodulate FM signals we will consider zero crossing detector. A description of the demodulation process for FM signals is illustrated in Figure 2.2.4.

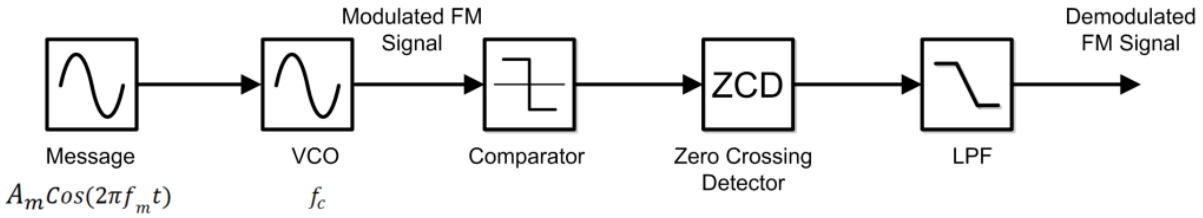


Figure 2.2.4. Demodulation of a FM signal

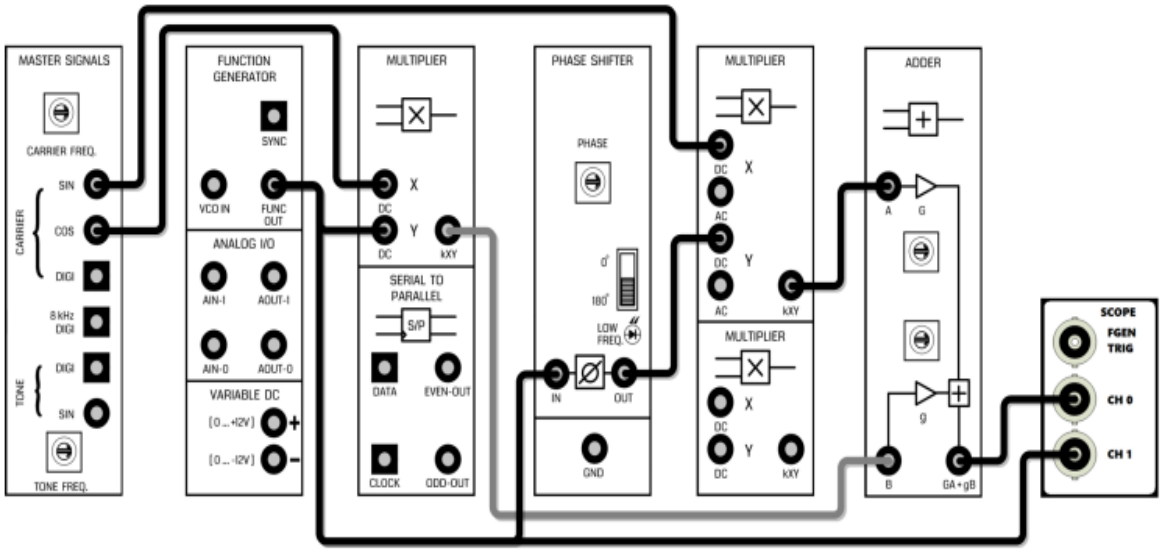
3. PROCEDURE

3.1. Generating Single Side Band (SSB) Signal

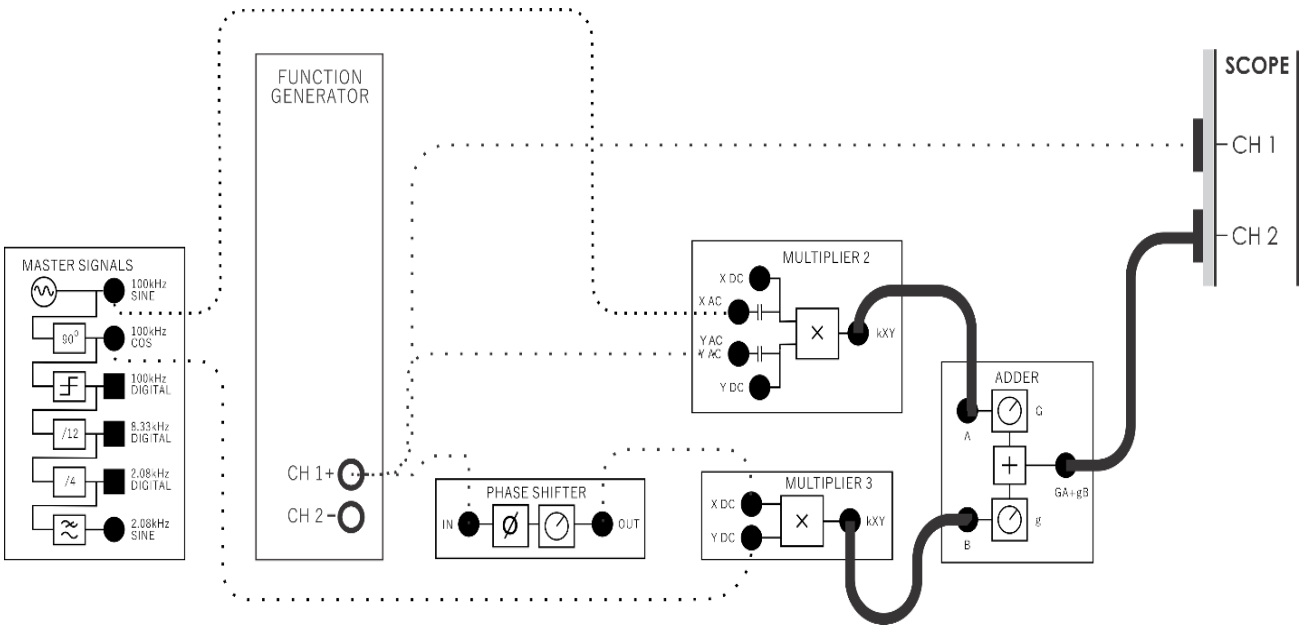
Red Set	Blue Set
• Connect the NI ELVIS platform to PC via USB. Turn on the power of NI ELVIS platform and TETRA-BB board. • Set the control mode of TETRA-BB to PC mode by using the CONTROL MODE switch on the board. • Run the TETRA-BB Soft Front Panel (SFP). • Select the "Exp. 3 - Double Side Band and Single Side Band Modulation & Demodulation" experiment from the menu, TETRA-BB SFP will provide guidance for the steps of implementing the process of SSB modulation, whose block diagram and the required TETRA-BB connections are shown in Figure 3.1-a. • Set the carrier frequency (fc) to 100kHz. Set the message frequency fm to 2 kHz and peak to peak amplitude to Vpp = 4 V of the message signal m(t). • Adjust the phase shifter for 90° (or 270°) using the TETRA-BB SFP phase shifter block. Set input signal frequency in the phase shifter block. • Set gain parameters g and G to 1 of the adder block. • Adjust following the oscilloscope panel: Channel 0 Voltage Range: 2V/div Channel 1 Voltage Range: 2V/div Horizontal Time base: 100µs/div Trigger: Immediate	▪ Ensure that the NI ELVIS III Application Board power button at the top left corner of the unit is OFF (not illuminated). ▪ Carefully plug the Emona Communications board into the NI ELVIS III ensuring that it is fully engaged both front and back. ▪ Ensure that you have connected the NI ELVIS III to the PC using the USB cable and that the PC is turned on. ▪ Turn on the Application Board *Power* button by pressing it once and confirm that it is illuminated. The LEDs on the board should also be illuminated. If they are not, then switch the unit off immediately and check for connection or insertion errors. ▪ Open the Instrument Launcher software in your browser and select the required instruments. ▪ Adjust following Scope configurations: Channel 1 Voltage Range: 2V/div Channel 2 Voltage Range: 2V/div Horizontal Time base: 50µs/div Trigger: Analog Edge, Chan 1, Rising Probe Attenuation :1x ▪ Modify the set-up as shown in figure 3.1-b. ▪ Set the message frequency fm to 2 kHz and peak to peak amplitude to Vpp = 4 V of the message signal m(t) on the Function Generator. ▪ Set the Phase Shifter module's *Phase Adjust* control to about the middle of its travel. ▪ Locate the Adder module and set BOTH its *G* and *g* controls to about the

	middle of their travel. This is about unity gain.

1. Observe the modulated and demodulated signals  .



(a)



(b)

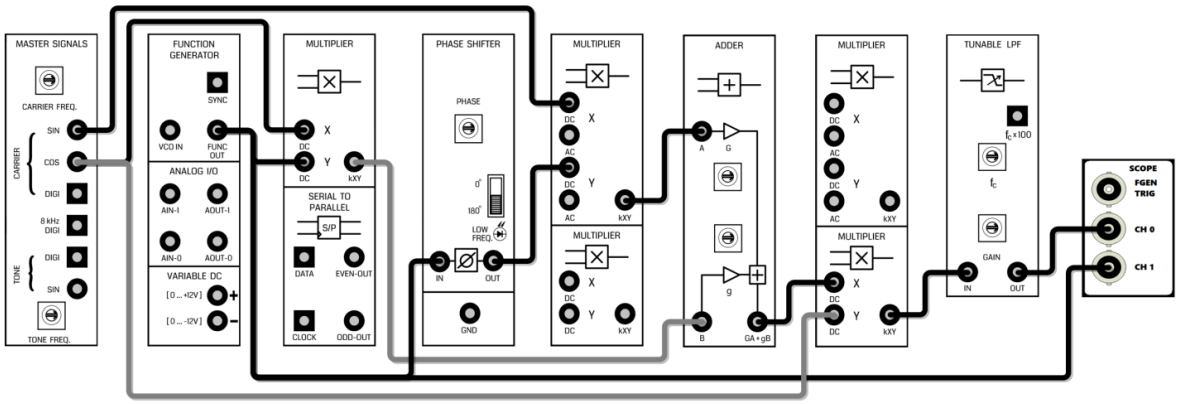
Figure 3.1. Wiring of (a) TETRA-BB (b) EMONAS hardware blocks to generate an SSB signal

Red Set	Blue Set
• Stop oscilloscope. • Switch from the oscilloscope window/mode to the DSA panel. • Apply the settings of DSA given below: Input Settings: Channel 0 Center Frequency: 100000 Frequency Span: 60000 Mode: RMS Resolution (lines): 1600 Weighting: Exponential Window: 7 Term-B Harris # of Averages: 5	▪ Enable the FFT mode of the Oscilloscope instrument. Change the scope's timebase to 1ms/div. ▪ Set the frequency span for the FFT displayed from say 80kHz to 120kHz for a closeup of the frequency domain of interest. ▪ Select the Hamming window, source channel to Channel 2.

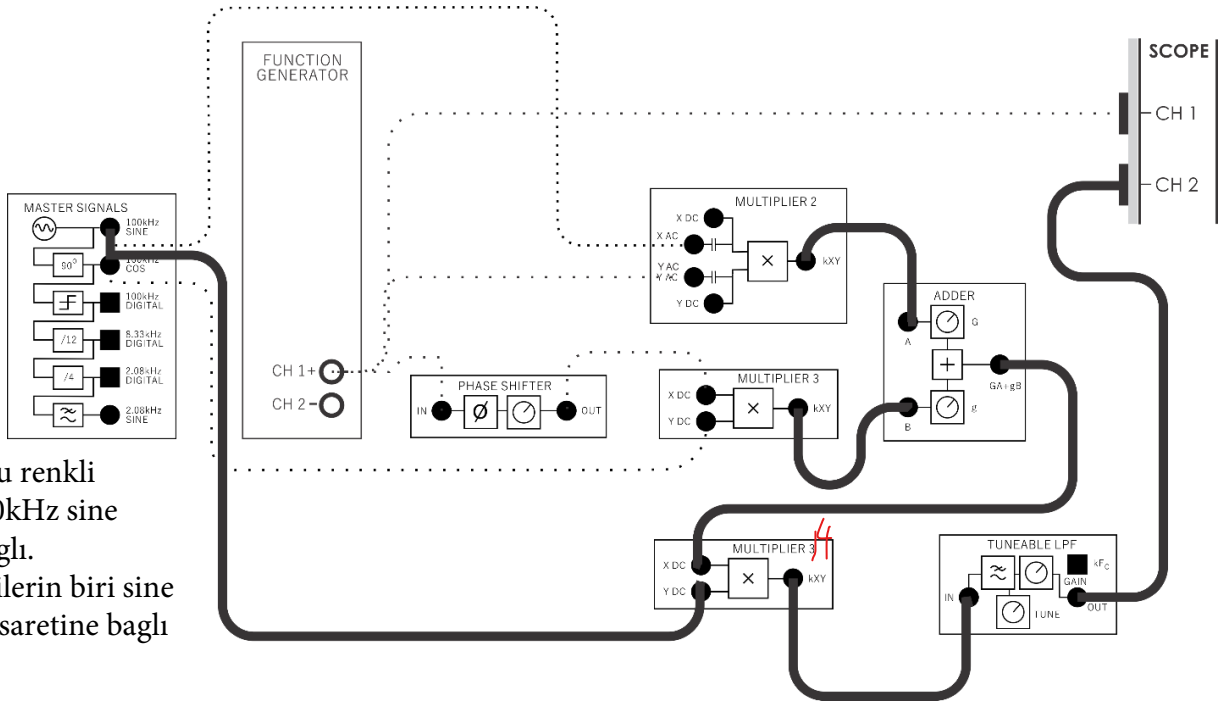
2. Observe the frequency spectrum of the modulated signal. Then, measure frequencies and amplitudes of each frequency components. 📷,✎ Stop DSA and Swich to oscilloscope in SPF.

3.2. Demodulating an SSB Signal

Red Set	Blue Set
• Select "Exp. 3 - Double Side Band and Single Side Band Modulation & Demodulation" experiment from the menu, TETRA-BB software will provide guidance for the steps of implementing the process of DSB demodulation, whose block diagram and the required TETRA-BB connections are shown in Figure 3.2-a. • Set the gain of tunable low pass filter to 1. Then, adjust the cut-off frequency of tunable low pass filter to 5kHz.	▪ Locate the Tuneable Low-pass Filter module on the board and set its *Gain* control to about the middle of its travel. ▪ Turn the Tuneable Low-pass Filter module's *Cut-off Frequency Adjust* control fully clockwise. ▪ Modify the set-up as shown in Figure 3.2-b.



(a)



Burada koyu renkli bağlantı 100kHz sine isaretine baglı. Kesikli çizgilerin biri sine biri cosine isaretine baglı

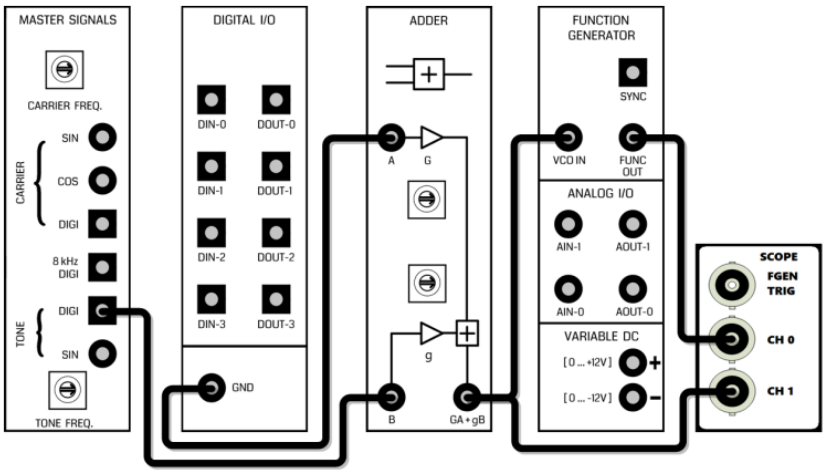
(b)

Figure 3.2. Wiring of (a) TETRA-BB (b) EMONAS hardware blocks to SSB demodulator

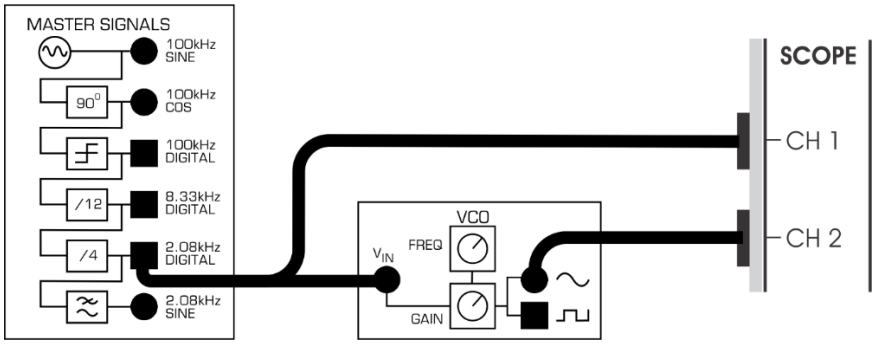
1. Observe and draw the message signal m(t) and the demodulated signal ySSB(t) 📷. Measure the amplitude and frequency of the demodulated signal ySSB(t). Compare the frequencies of demodulated signal and message signal. Then, discuss these results✒.

3.3. Generating an FM Signal

Red Set	Blue Set
• Connect the NI ELVIS platform to PC via USB. Turn on the power of NI ELVIS platform and TETRA-BB board. • Set the control mode of TETRA-BB to PC mode by using the CONTROL MODE switch on the board. • Run the TETRA-BB Soft Front Panel (SFP). • Select the e "Exp. 4 - Frequency Modulation & Demodulation" experiment from the menu, TETRA-BB SFP will provide guidance for the steps of implementing FM modulator, whose block diagram and the required TETRA-BB connections are shown in Figure 3.3-a. • Set peak to peak amplitude of the sinus signal in Function Generator to Vpp = 4 V and 50 kHz. Then, set digital tone signal frequency fm to 2 kHz of the message signal m(t) in the Master Signal block. • Set the “Modulation Type” to FM in function generator. • Set the gain (g) of adder to 0.3. • Adjust the voltage ranges and the time base on the oscilloscope panel so that you can clearly see the signals you have obtained.	▪ Ensure that the NI ELVIS III Application Board power button at the top left corner of the unit is OFF (not illuminated). ▪ Carefully plug the Emona Communications board (ECB) into the NI ELVIS III ensuring that it is fully engaged both front and back. ▪ Ensure that you have connected the NI ELVIS III to the PC using the USB cable and that the PC is turned on. ▪ Turn on the Application Board *Power* button by pressing it once and confirm that it is illuminated. The LEDs on the ECB should also be illuminated. If they are not, then switch the unit off immediately and check for connection or insertion errors. ▪ Open the Instrument Launcher software in your browser and select the required instruments. ▪ Adjust following Scope configurations: Channel 1 Voltage Range: 2V/div Channel 2 Voltage Range: 2V/div Horizontal Time base: 50µs/div Trigger: Analog Edge, Chan 1, Rising Probe Attenuation :1x ▪ Connect the set-up shown in Figure 3.3-b. ▪ Set the VCO module’s *GAIN* to maximum (fully clockwise) and then set the *FREQ* control to give an output sinusoid of about 100kHz. This has a period of 10us, so you can easily fine tune it in the time domain.



(a)



(b)

Figure 3.3. Wiring of (a) TETRA-BB (b) EMONAS hardware blocks to FM modulator

1. Observe the modulated and demodulated signals 📷,✎. (Adjust trigger settings, if you do not see the signal in oscilloscope.)
2. Then, disconnect digital tone signal from adder block. Connect sinus tone of the Master Signals to adder block ($g = 1$ for red sets). Observe the modulated signal 📷, ✎.

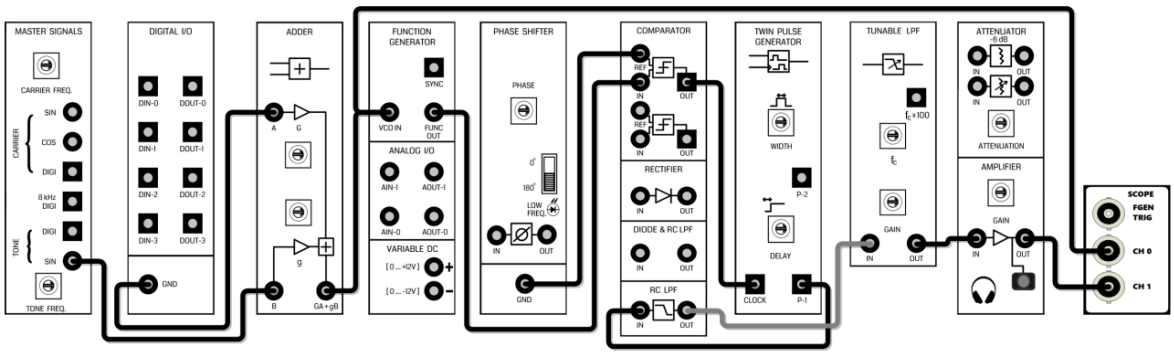
Red Set	Blue Set
• Stop oscilloscope. • Switch from the oscilloscope window/mode to the DSA panel. • Apply the settings of DSA given below: Input Settings: Channel 1 Center Frequency: 50000 Frequency Span: 100000 Mode: RMS Resolution (lines): 1600 Weighting: Exponential	▪ Enable the FFT mode of the Oscilloscope instrument. Change the scope's timebase to 1ms/div. ▪ Set the frequency span for the FFT displayed from say 0kHz to 200kHz for a closeup of the frequency domain of interest. ▪ Select the Hamming window, source channel to Channel 2.

Window: 7 Term-B Harris • # of Averages: 5	

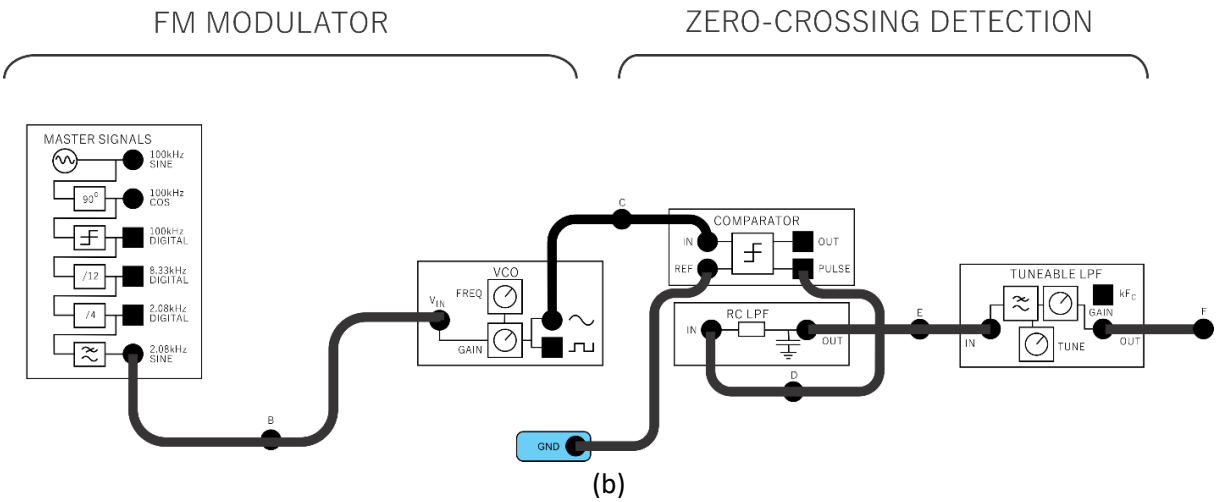
3. Observe and measure frequency components of the modulated signal for both digital and sinus tone signals. 📷,✎.
4. Find the (%90) bandwidths using the frequency spectrums 📷, ✎.

3.4. Demodulating an FM Signal

Red Set	Blue Set
• Select "Exp. 4 - Frequency Modulation & Demodulation" experiment from the menu, TETRA-BB software will provide guidance for the steps of implementing the process of DSB demodulation, whose block diagram and the required TETRA-BB connections are shown in Figure 3.4-a. • Set the g=0.3 and fm=5kHz • Adjust the cut-off frequency of the tunable low pass filter to re-generate the message signal at the output of the filter (7,89kHz, gain is 2). • Adjust the following settings: Twin Pulse Generator: width:5μs Delay: 405μs Amplifier gain: 7	▪ Turn the Tuneable Low-pass Filter module's *Gain* control fully clockwise. ▪ Turn the Tuneable Low-pass Filter module's *Cut-off Frequency Adjust* control fully clockwise. ▪ Modify the set-up as shown in Figure 3.2-b. ▪ Make the following adjustments to the scope's controls: Voltage per division: Channel 1 to *1V/div* and Channel 2 to *20mV/div* *Input Coupling* control for both channels to *AC* *Trigger Type* to *Analog Edge* *Trigger Source* to *Channel 1* *Timebase* control to *200µs/div*



(a)



(b)

Figure 3.2. Wiring of (a) TETRA-BB (b) EMONAS hardware blocks to FM demodulator

1. Observe and draw message signal and demodulated signal in time domain 📷, ✏.